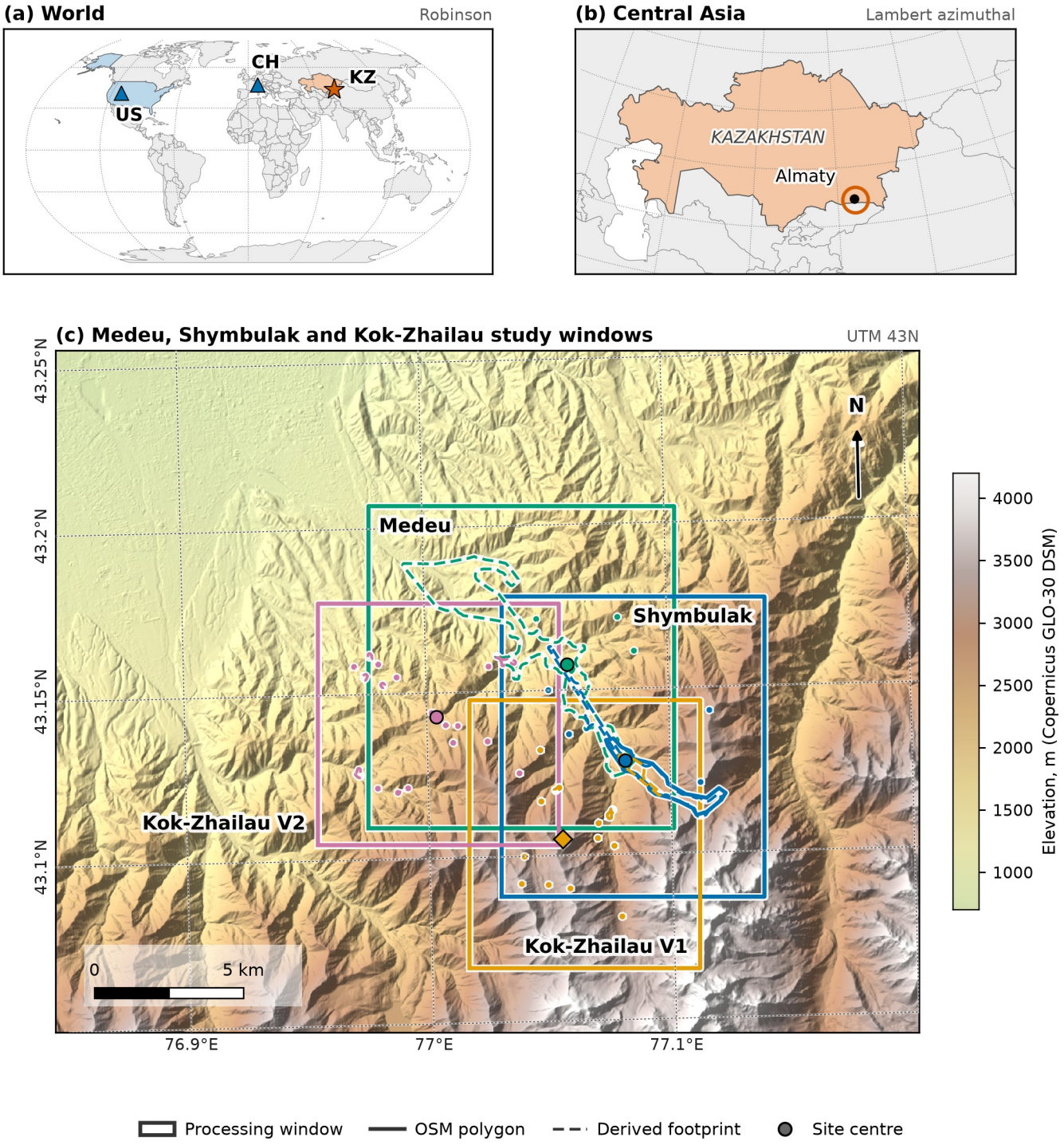


Fig A study area. Study area. (a) Location of the Kazakhstan study sites (KZ) and of the Swiss (CH) and US lidar validation regions. (b) Central Asia with the enlarged window. (c) Medeu, Shymbulak and two alternative Kok-Zhailau study windows (V1, V2) on Copernicus GLO-30 relief, 76.85-77.20 deg E, 43.05-43.25 deg N, UTM zone 43N. Coloured rectangles are analysis windows, not administrative borders. Solid lines are OpenStreetMap polygons and dashed lines are footprints derived from mapped objects; neither is an official boundary.

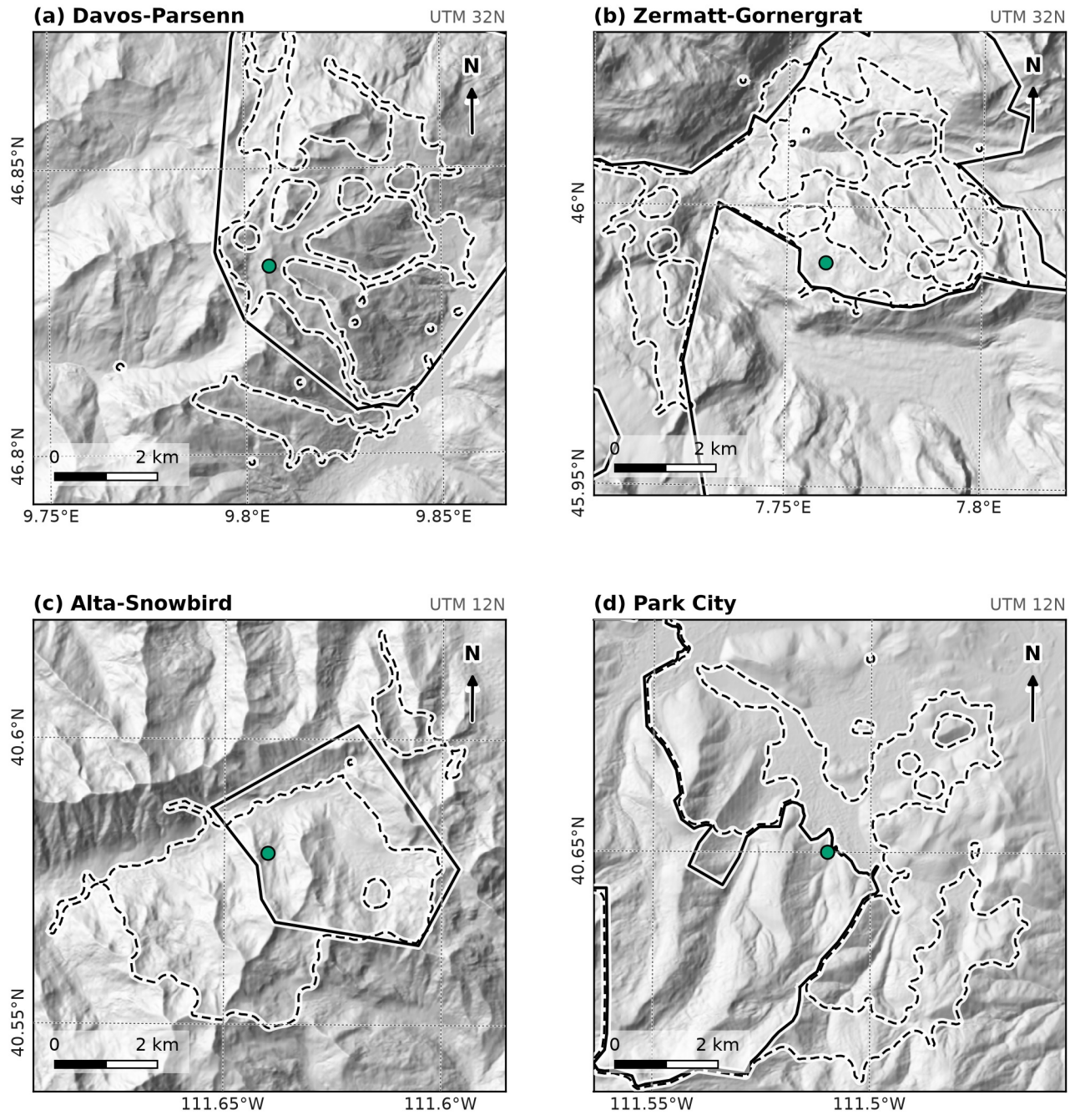


Fig A suppl validation sites. Lidar validation sites. Davos-Parsenn and Zermatt-Gornergrat (Switzerland) and Alta-Snowbird and Park City (USA), each in its own UTM zone. Solid lines are OpenStreetMap polygons where one was found and dashed lines are footprints derived from mapped objects; neither is an official boundary.

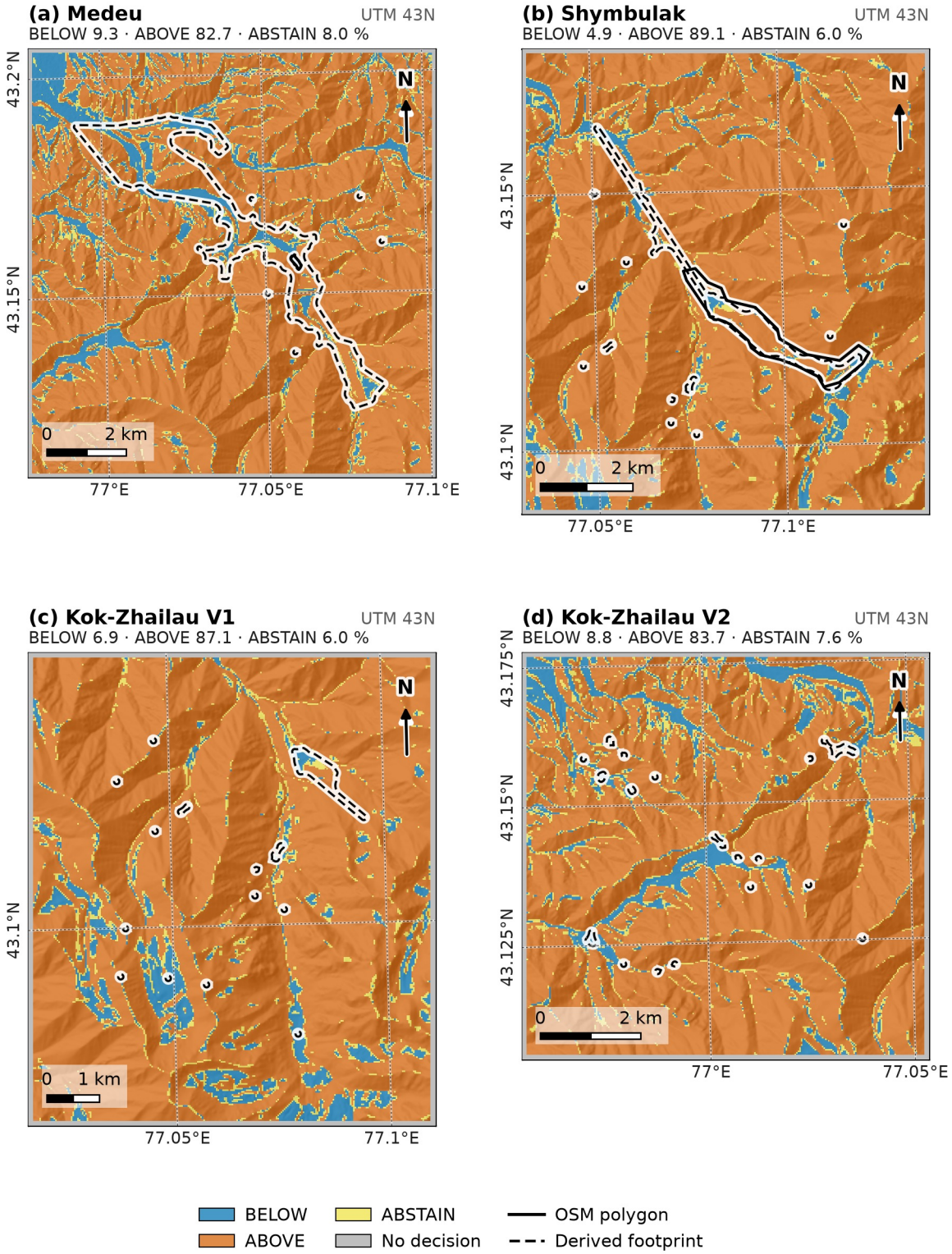


Fig B decisions 15deg. Threshold decisions at 15 deg slope for the four Kazakhstan study windows (Medeu, Shymbulak, Kok-Zhailau V1 and V2). Colours show the classes of the combined abstention rule (BELOW, ABOVE, ABSTAIN); grey = no decision. The rule was trained on lidar-validated sites in Switzerland and the USA and is transferred here, so the error is not measured at these sites. Percentages are shares of valid 30 m pixels per window. Outlines are exploratory and not official boundaries.

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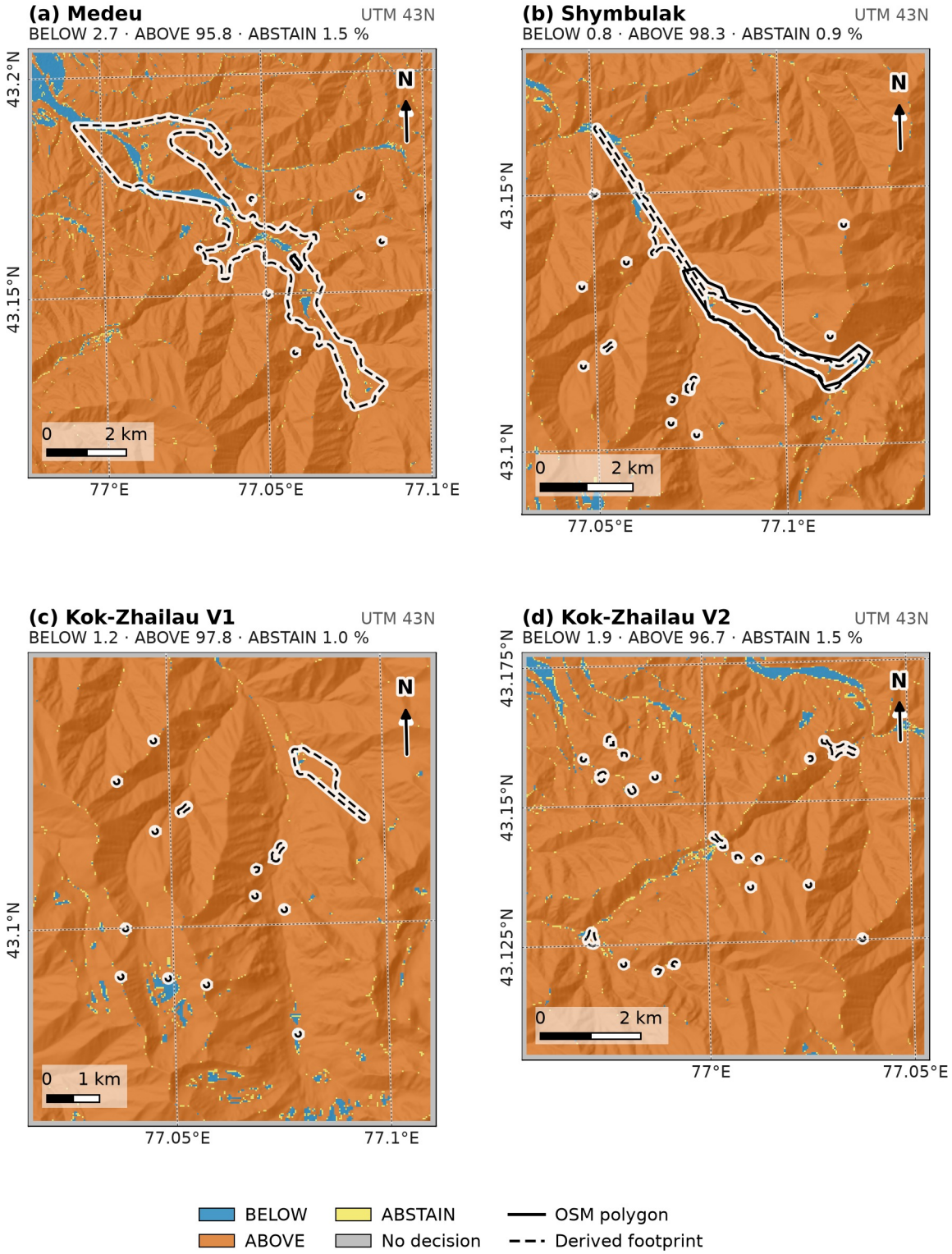


Fig B suppl decisions 7.5deg. Threshold decisions at 7.5 deg slope for the four Kazakhstan study windows (Medeu, Shymbulak, Kok-Zhailau V1 and V2). Colours show the classes of the combined abstention rule (BELOW, ABOVE, ABSTAIN); grey = no decision. The rule was trained on lidar-validated sites in Switzerland and the USA and is transferred here, so the error is not measured at these sites. Percentages are shares of valid 30 m pixels per window. Outlines are exploratory and not official boundaries.

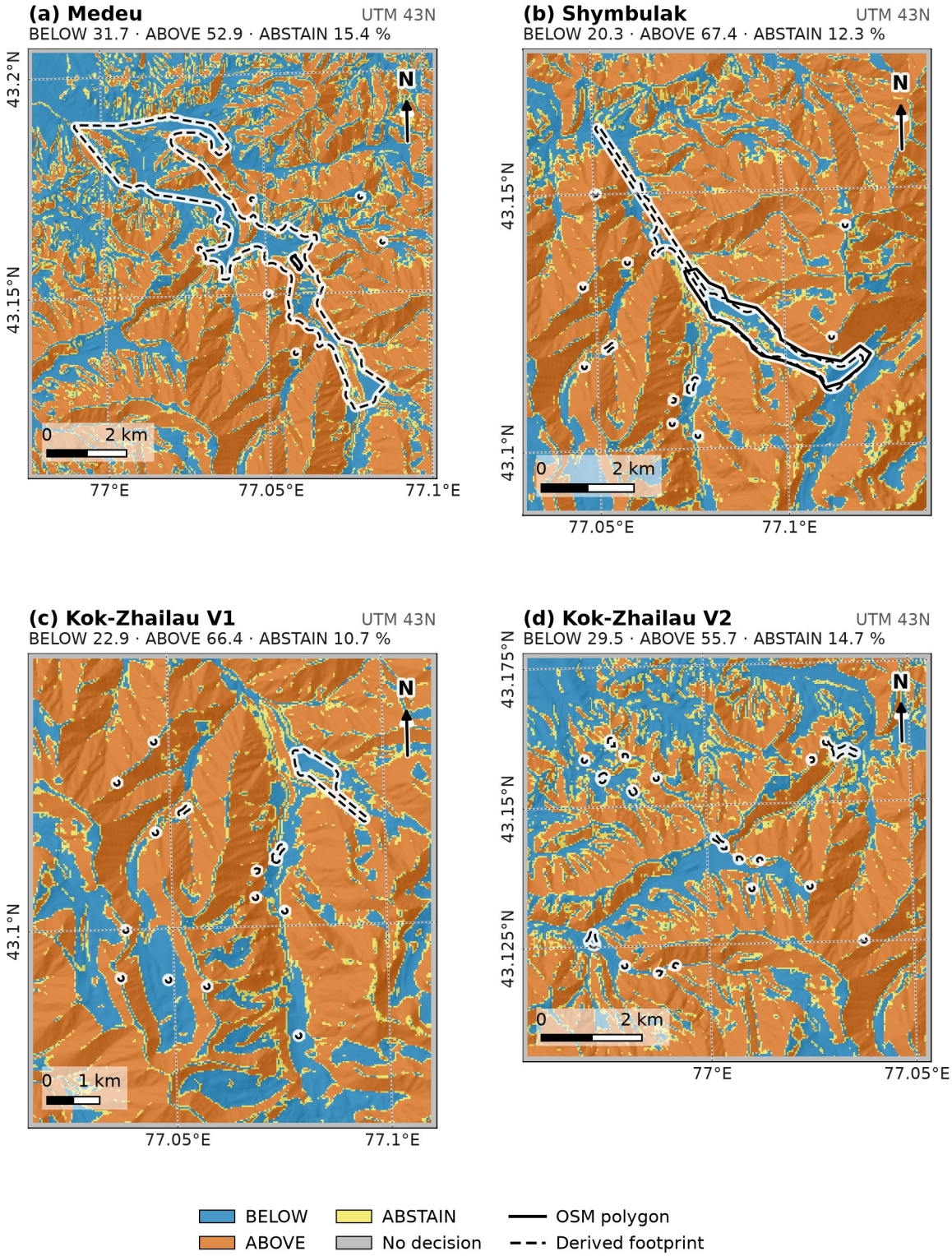


Fig B suppl decisions 25deg. Threshold decisions at 25 deg slope for the four Kazakhstan study windows (Medeu, Shymbulak, Kok-Zhailau V1 and V2). Colours show the classes of the combined abstention rule (BELOW, ABOVE, ABSTAIN); grey = no decision. The rule was trained on lidar-validated sites in Switzerland and the USA and is transferred here, so the error is not measured at these sites. Percentages are shares of valid 30 m pixels per window. Outlines are exploratory and not official boundaries.

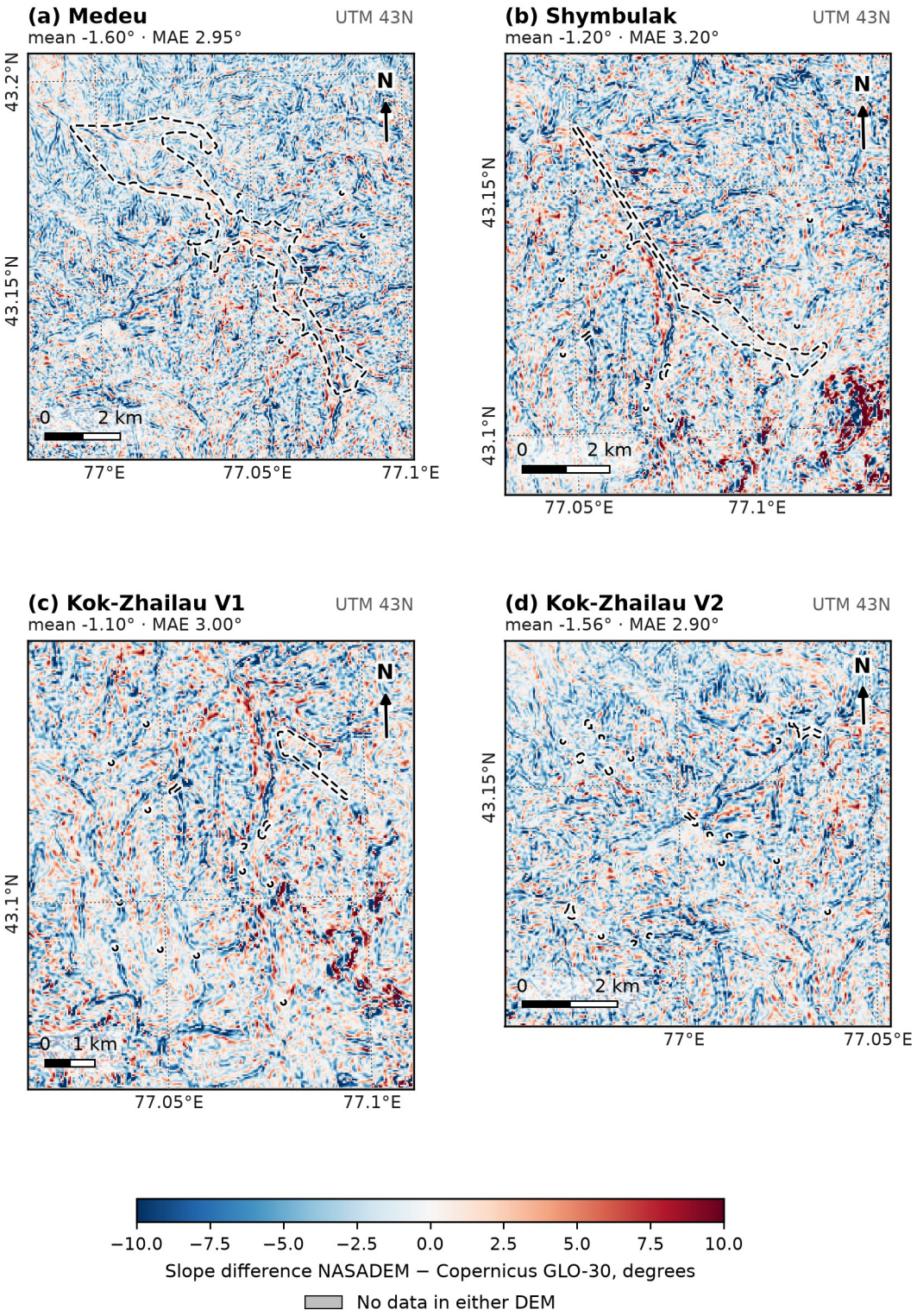


Fig C1 kazakhstan inter dem slope difference. Slope differences between two global DEMs (NASADEM minus Copernicus GLO-30) in the four Kazakhstan windows, on the common valid mask. The map shows disagreement between DEMs, not accuracy: no lidar reference exists for Kazakhstan. Dashed line = derived footprint (exploratory).

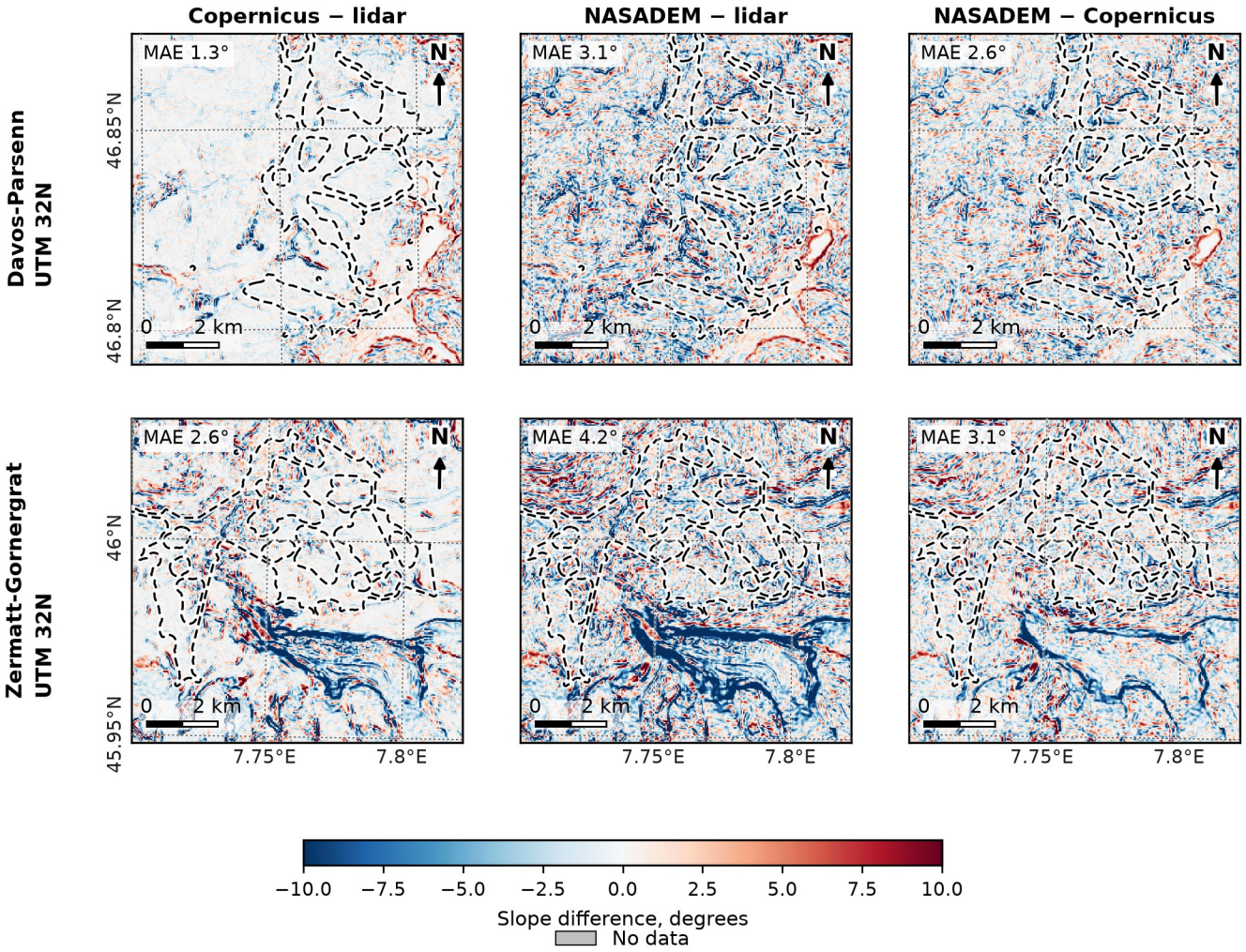


Fig C2 switzerland dem vs lidar slope difference. Slope differences at the lidar validation sites in Switzerland: global DEM minus lidar-DTM reference and NASADEM minus Copernicus GLO-30, on the common valid mask of each pair. The lidar DTM is block-averaged to the 30 m cell before the slope is computed (support-matched). MAE is the mean absolute difference.

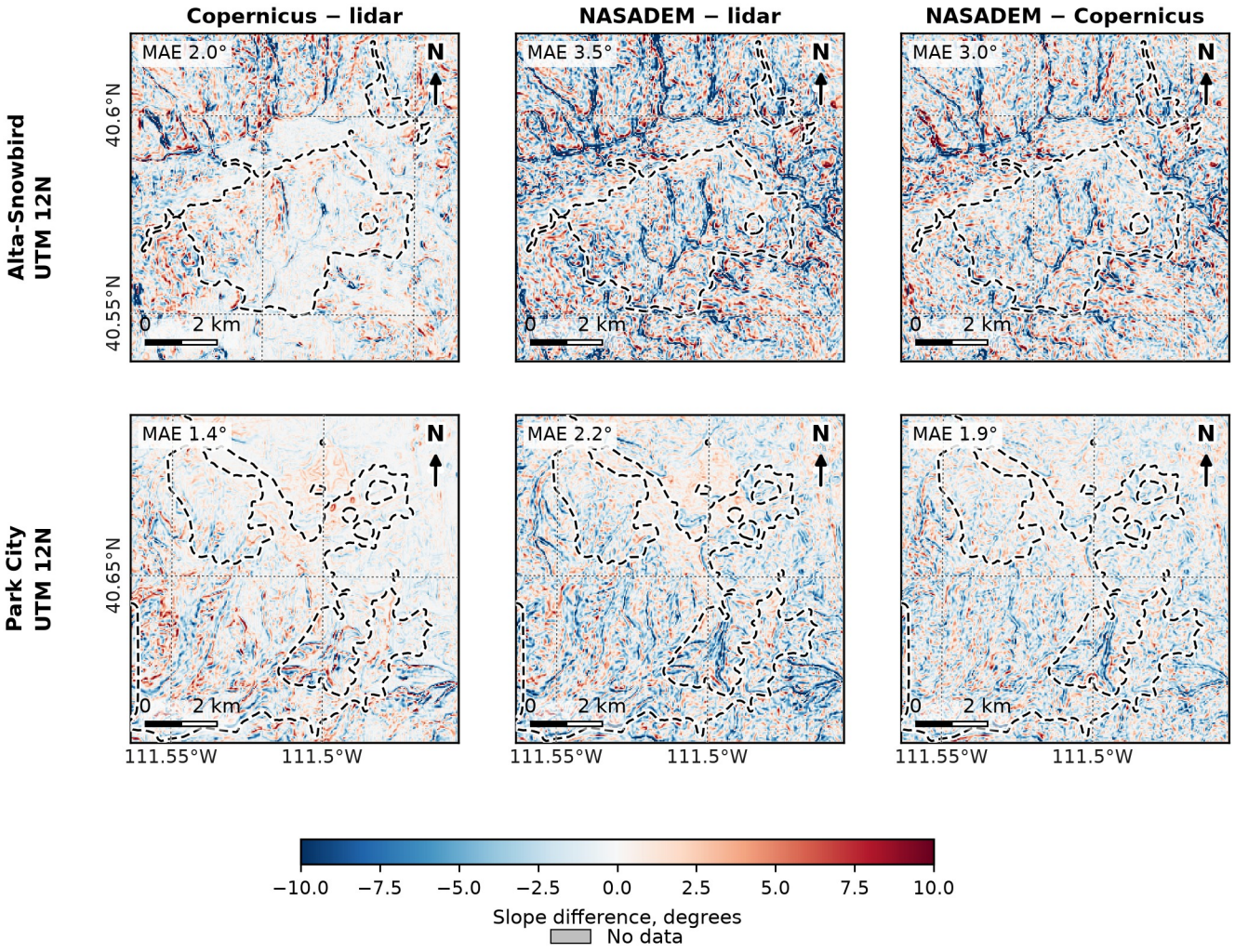


Fig C2 usa dem vs lidar slope difference. Slope differences at the lidar validation sites in USA: global DEM minus lidar-DTM reference and NASADEM minus Copernicus GLO-30, on the common valid mask of each pair. The lidar DTM is block-averaged to the 30 m cell before the slope is computed (support-matched). MAE is the mean absolute difference.

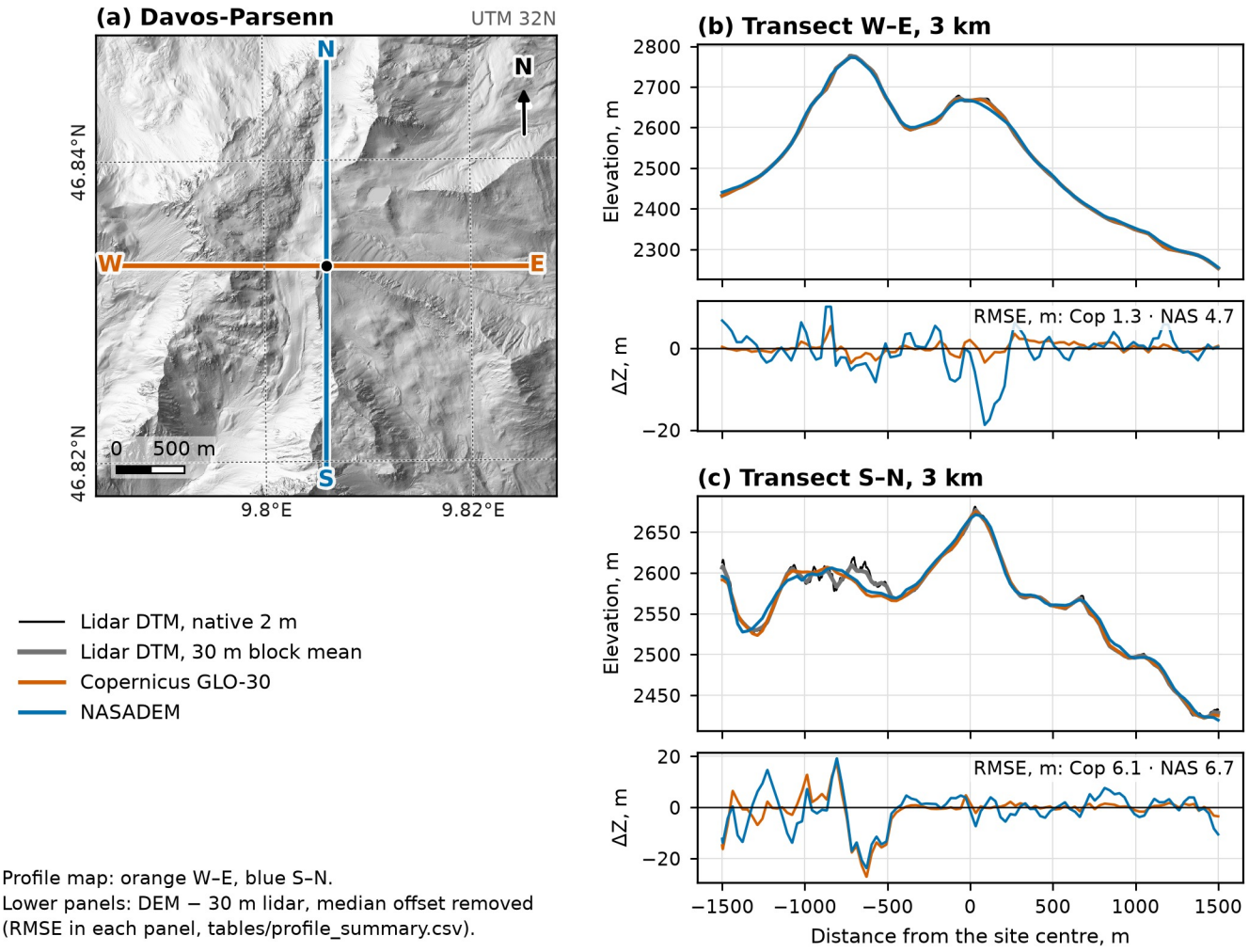
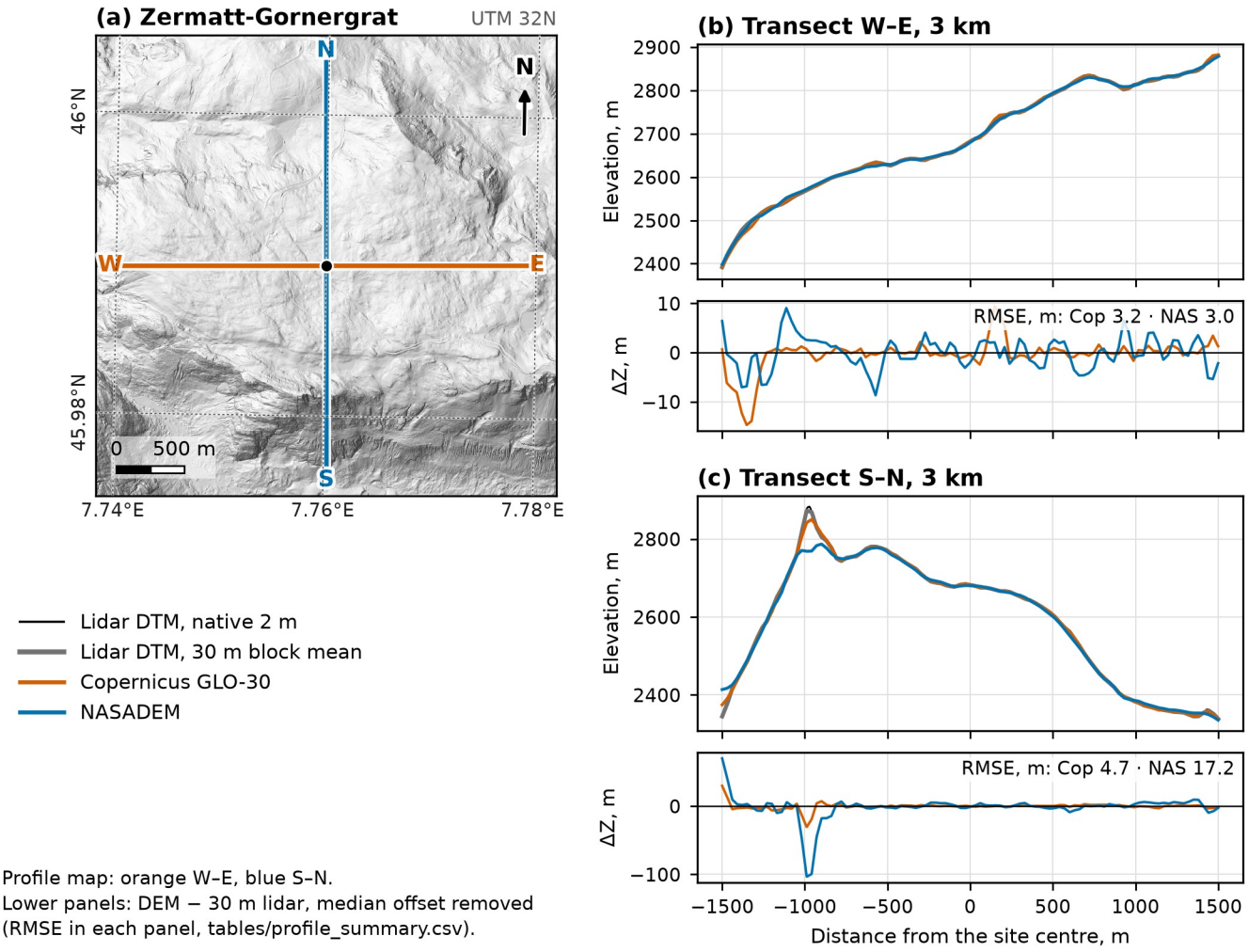
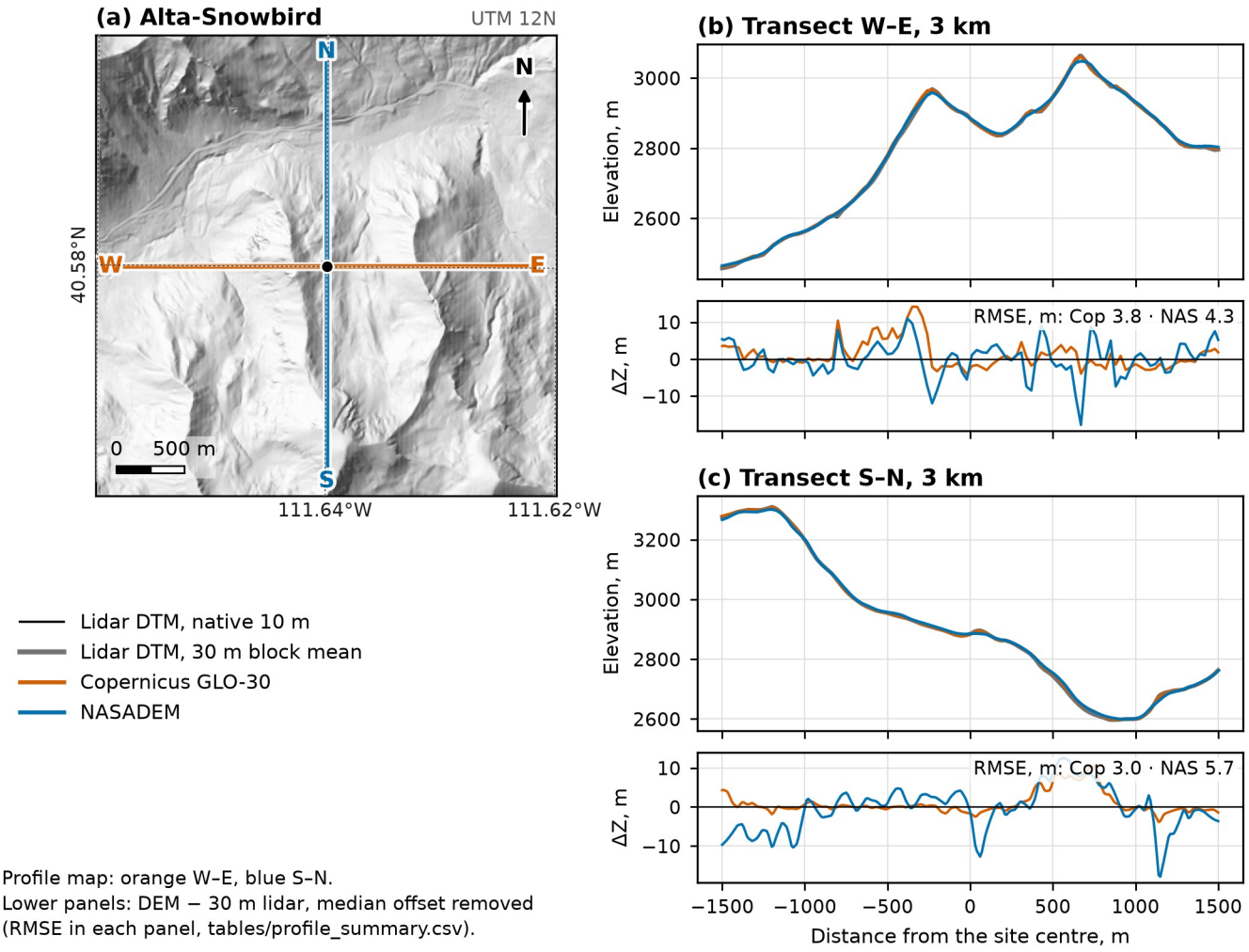


Fig D profile davos parsenn. Measured terrain profiles at Davos-Parsenn: two 3 km grid-aligned transects through the site centre (W–E, S–N) from the lidar DTM at native resolution, the same DTM averaged to the 30 m cell (support-matched reference) and the two global DSMs. Lower panels show DEM minus 30 m reference after removing the median offset.



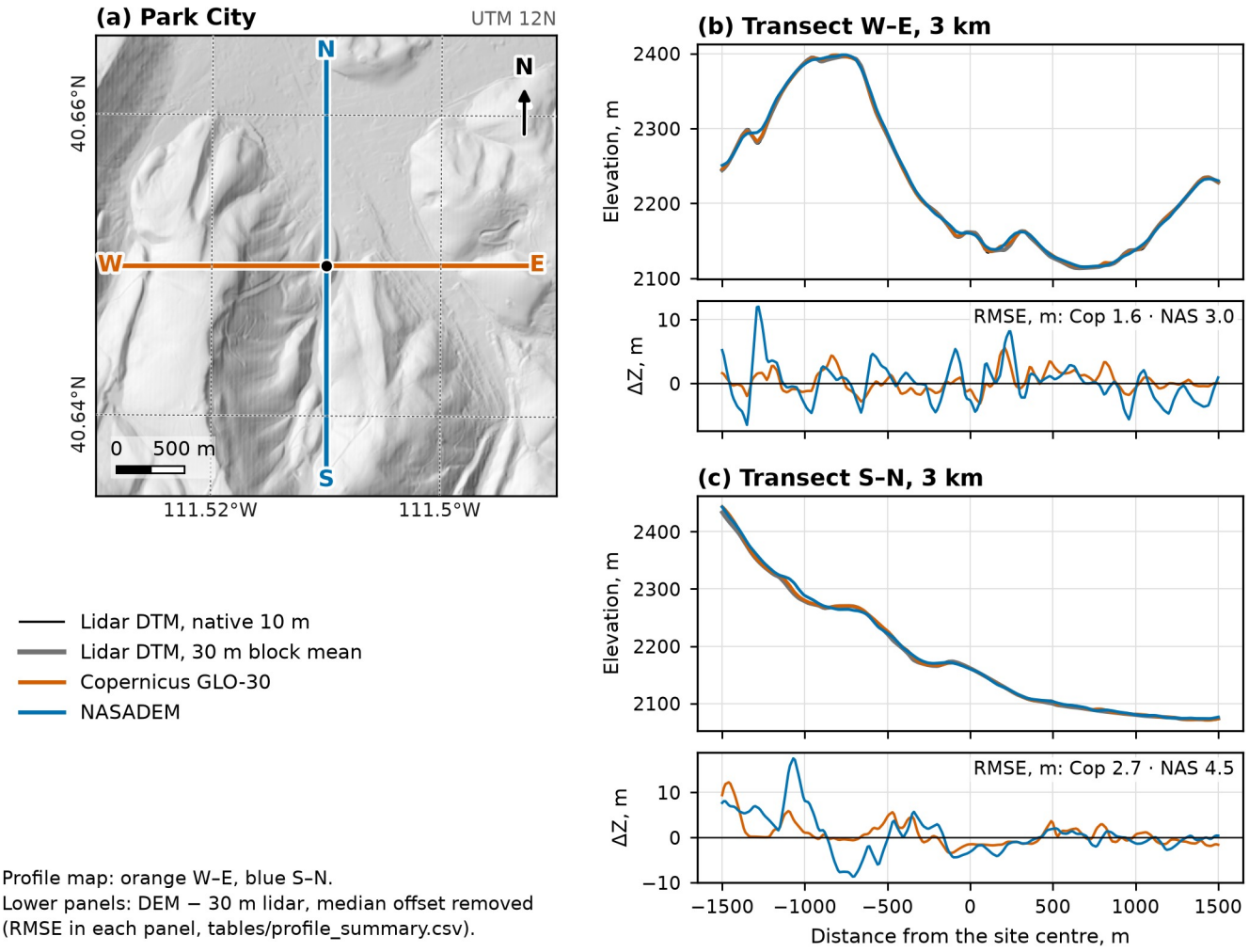
Profile map: orange W–E, blue S–N.
Lower panels: DEM − 30 m lidar, median offset removed
(RMSE in each panel, tables/profile_summary.csv).

Fig D profile zermatt gornergrat. Measured terrain profiles at Zermatt-Gornergrat: two 3 km grid-aligned transects through the site centre (W–E, S–N) from the lidar DTM at native resolution, the same DTM averaged to the 30 m cell (support-matched reference) and the two global DSMs. Lower panels show DEM minus 30 m reference after removing the median offset.



Profile map: orange W–E, blue S–N.
Lower panels: DEM − 30 m lidar, median offset removed
(RMSE in each panel, tables/profile_summary.csv).

Fig D profile alta snowbird. Measured terrain profiles at Alta-Snowbird: two 3 km grid-aligned transects through the site centre (W–E, S–N) from the lidar DTM at native resolution, the same DTM averaged to the 30 m cell (support-matched reference) and the two global DSMs. Lower panels show DEM minus 30 m reference after removing the median offset.



Profile map: orange W–E, blue S–N.
Lower panels: DEM − 30 m lidar, median offset removed
(RMSE in each panel, tables/profile_summary.csv).

Fig D profile park city. Measured terrain profiles at Park City: two 3 km grid-aligned transects through the site centre (W–E, S–N) from the lidar DTM at native resolution, the same DTM averaged to the 30 m cell (support-matched reference) and the two global DSMs. Lower panels show DEM minus 30 m reference after removing the median offset.

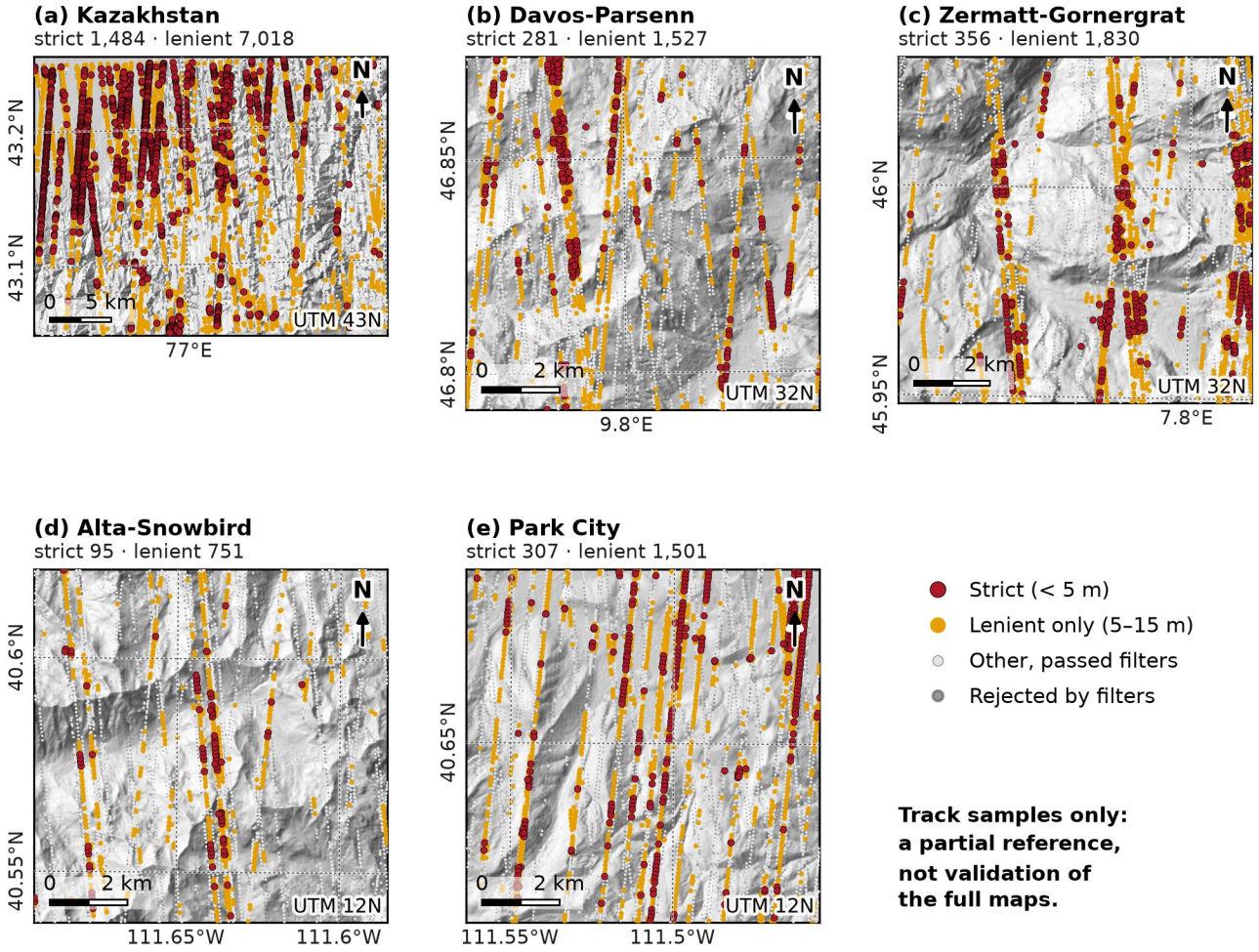


Fig E atl08 track samples. Real ICESat-2 ATL08 terrain segments (100 m along-track) over the Kazakhstan window and the four lidar validation sites. Strict (uncertainty < 5 m) and lenient (< 15 m) subsets are shown on top of the other segments. Segments exist only along ground tracks; they are a partial independent reference and must not be read as validation of any full slope map.